

Data Science Masters

Numpy Basic — Assignment Answers

Q1. Is there any difference in the data type of variables `list_` and `array_list`?

Yes. `list_` is a Python list, while `array_list` is a NumPy ndarray. The elements of `list_` are strings, and NumPy creates `array_list` with a string data type as well.

```
import numpy as np

list_ = ['1', '2', '3', '4', '5']
array_list = np.array(object=list_)

print(type(list_))
print(type(array_list))
print(list_[0].__class__)
print(array_list.dtype)
```

Q2. Write a code to print the data type of each and every element of both variables.

The elements of `list_` are Python strings. The elements accessed from the NumPy array have NumPy's string type.

```
print("Data types of list_ elements:")
for x in list_:
    print(type(x))

print("\nData types of array_list elements:")
for x in array_list:
    print(type(x))
```

Q3. Considering `array_list = np.array(object=list_, dtype=int)`, will there be any difference in the data type of the elements?

Yes. `list_` still contains strings because it was not changed. By specifying `dtype=int`, the NumPy array converts the string values into integers. Therefore, the list elements are strings while the NumPy array elements are integers.

```
array_list = np.array(object=list_, dtype=int)

print("Data types of list_ elements:")
for x in list_:
    print(type(x))

print("\nData types of array_list elements:")
for x in array_list:
    print(type(x))
```

Q4. Write a code to find the shape and size of `num_array`.

For the given nested list, the NumPy array has 2 rows and 3 columns, so its shape is (2, 3). It contains 6 elements, so its size is 6.

```
num_list = [[1, 2, 3], [4, 5, 6]]
num_array = np.array(object=num_list)

print("Shape:", num_array.shape)
print("Size:", num_array.size)
```

Q5. Create a NumPy array of a 3×3 matrix containing zeros only.

Use NumPy's `zeros()` function. The resulting array has shape (3, 3) and contains 9 zeros.

```
zero_array = np.zeros((3, 3))

print(zero_array)
print("Shape:", zero_array.shape)
print("Size:", zero_array.size)
```

Q6. Create an identity matrix of shape (5,5) using NumPy functions.

Use NumPy's `eye()` function. It creates a 5×5 matrix with 1s on the main diagonal and 0s elsewhere.

```
identity_matrix = np.eye(5)

print(identity_matrix)
```

Note: The uploaded assignment asks for the work to be created in a Jupyter Notebook and uploaded to a public GitHub repository, with the repository link shared through the dashboard.